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SUMMARY

AI engineer and empirical safety researcher who builds systems that report honestly on their own uncertainty. I ship agentic LLM products with adversarial input screening and grounding validation, run evaluation and interpretability studies on open models (CoT faithfulness, sycophancy, deception probes, SAEs and attribution graphs), and write the security and infrastructure layer underneath both. M.S. Cybersecurity, Penn State; CCNA and Security+; F-1 OPT work authorized.

EXPERIENCE

Researcher, AI Sycophancy · The Pennsylvania State University · State College, PA 2026 – Present

- Reverse-engineering sycophancy in Gemma-3-4B, defined as a verdict that changes under a transformation that should not change it, so each scenario is its own control.
- Working bottom-up through eleven gated rungs, from attention patterns to causal validation by steering and ablation.
- Worded every claim to the evidence supporting it, reserving *is the mechanism for* to patching with specificity; null results are written up with equal care.

AI/ML Engineer · Lokesh Agro Foods · Chennai, India 2025 – Jun 2026

- Rearchitected a hardcoded classify/search/format pipeline into a Gemini 2.0 Flash tool-calling agent that chooses tool order per conversation.
- Built a defense-in-depth path: a local Llama-3-8B injection screen before tool routing, server-side authorization on lookups, and a grounding check before send.
- Grounded generation on ChromaDB with section-level chunking, so answers carry citations and the agent escalates rather than guessing.
- Split a Node orchestrator from a Python safety/RAG service, keeping the grounding layer testable and the channel interchangeable.
- Built produce-grading image classification and a demand model, replacing spreadsheet-driven inventory planning.
- Designed the wellness assistant with explicit non-diagnostic boundaries and transparent sourcing for every recommendation.

Infrastructure Security Engineer Intern · iMerit · Remote Jun 2025 – Aug 2025

- Scoped identity integration across Ivanti ITSM, Google Workspace, and Azure AD to unify onboarding across three disconnected systems.
- Defined role boundaries and access-control requirements for the provisioning path, tightening new-hire onboarding.

Graduate Teaching Assistant · The Pennsylvania State University · State College, PA Aug 2024 – May 2026

- Taught Java and Haskell to 100+ graduate students, focused on correctness, invariants, and asymptotic complexity.
- Fed recurring failure patterns back to instructors to reshape problem sets around measured weak points.

Platform Engineer · Tvastr · Chennai, India Jul 2022 – Dec 2022

- Built a reactive AWS governance engine flagging compliance drift across a 1000+ account estate.

AI Research Intern · Anna University (CEG) · Chennai, India Feb 2023 – Jul 2024

- Built a hybrid LSTM + 1D CNN intrusion-detection pipeline on UNSW-NB15 (2.5M flows, 9 attack classes) with SMOTE for minority classes.
- Implemented two-stage metaheuristic feature selection (Sine Cosine, then Particle Swarm), dropping false positives by 25%.

PROJECTS

Chain-of-Thought Faithfulness Evaluation

RESEARCH

Llama-3-8B-Instruct, vLLM, BIG-Bench-Hard, A100-80GB

- Replicated Turpin and Lanham on Llama-3-8B across 13 BIG-Bench-Hard tasks; 9,561 deterministic temp-0 calls through a four-stage vLLM pipeline.
- Designed two biasing protocols to separate format bias from sycophancy; the positional one alone collapsed date understanding from 66% to 19%.
- Measured 63.7% unfaithfulness against 36.3% articulation and a 12.3% mean accuracy drop.

- Found self-judging fails — the model's sycophancy reappeared in the judge seat — and replaced it with an independent judge.

Secret Loyalty Detection in Fine-Tuned Models

RESEARCH

Apart Research sprint, Qwen2.5-7B-Instruct

- Audited three Qwen2.5-7B-Instruct fine-tunes for covert weight-encoded loyalties across ~8,700 generations and 127 actors.
- Confirmed the null model four independent ways and localized real modification in the other two to layers 10–12 by activation divergence.
- Ran per-actor separability, affinity probing, and causal patching; every effect was global rather than targeted. Reported a negative result.
- Documented a failure mode: prefill elicitation extracted "confessions" from the known-clean base model at 22.9%.

Linear & Deception Probes on Transformer Internals

RESEARCH

PyTorch forward hooks, 7B models, residual stream

- Trained linear probes on residual-stream activations to read internal states the model's own text output denies, across three 7B models.

Attribution Graphs, SAEs & Cross-Layer Transcoders

RESEARCH

GPT-2 Small, sparse autoencoders, circuit analysis

- Implemented the interpretability ladder end to end, from linear probes through cross-layer transcoders and attribution graphs.
- Traced an indirect-object-identification circuit in GPT-2 and causally validated it by feature ablation.

LLM Sycophancy Evaluator

RESEARCH

Llama-3.2-3B, HuggingFace, Streamlit, Jinja2, pytest

- Built a paired-prompt harness for agreement bias, pressure capitulation, and flattery resistance across 30 adversarial prompt pairs.
- Found sycophancy varies by domain and that base models produce repetition loops inflating ambiguous scores. *Ongoing research.*

Second Brain — Goal-Driven Agent with Approval-Gated Execution

AI-SECURITY

Flower hackathon team build, Flower AgentApp, Python, Nebius Token Factory (Kimi K2.7, MiniMax M3), model2vec, WhatsApp, Google Calendar

- Built Second Brain, a Flower AgentApp that turns a goal into scored decision cards from four agents; nothing executes without an approved card.
- Treated every external text as data: tagged prompts, URL-grounded event extraction, a regex-only WhatsApp channel, and allow-listed edit lists.
- Hardened execution: briefs only for approved cards, single-lined so injected text cannot forge a task or remove the STOP before the final click; turns, messages, and concurrency bounded.
- Wrote a provider router with schema validation, retry, rule-based defaults, and circuit breaking, plus a four-rung embedding ladder that works with zero keys.

VecGate — Access Control for Vector Databases

AI-SECURITY

Python, PostgreSQL/pgvector, FastAPI, Docker, MIT

- Published an open-source RBAC layer enforcing facet-level and chunk-level permissions inside one SQL query, so unauthorized data is never retrieved.
- Built per-facet partial HNSW indexes and shipped SDK, CLI, and HTTP surfaces over one shared core, packaged to PyPI.

DDoS Detection & Explainable IDS

AI-SECURITY

M.S. Capstone, PyTorch, BiLSTM + 1D Residual CNN, Gemini API

- Reached 94.11% three-class accuracy across 700K+ flows with a BiLSTM and 1D residual CNN ensemble, reducing 324 columns to 86.
- Added confidence-based escalation: 85% auto-classifies while low-confidence predictions route to an analyst rather than being forced into a label.
- Added a Temporal CNN variant and a Gemini-backed layer turning raw detections into SOC alerts with a recommended mitigation.

Phishing & Voice-Scam Training Playground

AI-SECURITY

TypeScript, LLM-generated adversarial content

- Built a training environment generating live phishing and voice-scam scenarios rather than static examples.

Embedding Efficiency Research

AI-SECURITY

Autonomous research agent, PyTorch SDPA, RTX A4000

- Adapted an autonomous research-agent loop to Ampere (Flash Attention 3 to PyTorch SDPA), targeting the embedding layer at ~31% of baseline parameters.

SQ3R Read — Autonomous Study Agent

SOFTWARE

Chrome MV3, React, TypeScript, Node.js, SQLite, Vitest

- Architected a Chrome extension and Node companion turning any page into spaced-repetition cards via a five-stage LLM pipeline with FSRS-4.5 scheduling.
- Designed a finite-state batch queue over mutex-locked chrome.storage, with a service worker spawning review agents unsupervised.
- Built a pluggable provider layer over local Ollama and hosted APIs, under a rule that all model output is untrusted until validated.

Letterboxd Agent Memory — Mastra Workflow over Elasticsearch

SOFTWARE

TypeScript, Mastra, Elasticsearch Serverless, MCP

- Built a recommend → watch → review → post → re-ingest loop as a Mastra workflow, suspending across a film-length pause and a consent gate that agents cannot hold.
- Grounded recommendations on a Letterboxd corpus in Elasticsearch with recency-weighted recall, re-ingesting each review into the same index.
- Drove availability checks and posting through browser MCP tools, converting return-value errors into throws after a silent failure.

MarketMirror — Founder Idea Validator

SOFTWARE

Hackathon team build, TypeScript, Express, Exa, Gemini, Vapi/ElevenLabs

- Led the architecture of an idea validator that answers a pitch with cited Exa evidence, a skeptical Gemini persona, and a live Vapi voice call the founder must argue with.
- Built the Express validation and voice API, holding private provider keys server-side and out of the client-shipped variables that would leak them.

QUIC Router Simulation

SOFTWARE

C++17, discrete-event simulation, queue scheduling

- Wrote a packet-level discrete-event simulator comparing FIFO, WFQ, and HPFQ across three traffic classes, measuring P99 tail latency.

Self-Hosted Inference Stack

SOFTWARE

llama.cpp, CUDA, Python asyncio, Linux

- Ran a local inference stack with a benchmark suite measuring time-to-first-token, tokens/sec across batch sizes 1-8, and latency under load.
- Found memory bandwidth, not compute, is the ceiling, and that quantization loss is task-specific — negligible on summarization, widening on reasoning.

YAML Repository Memory Harness

SOFTWARE

YAML, filesystem index, agentic coding

- Built a YAML index over a codebase — every module named with its purpose, every folder described down to line ranges.
- Agents read the index rather than the source, keeping working context small and agentic coding fast.

Additional Systems Work

SOFTWARE

C++, FastAPI, React, PostgreSQL, Docker

- SSSEngine — zero-dependency C++ static site generator with incremental build caching; generates lkslovesh.com.
- HEVC streaming platform — FastAPI, React, and a Python worker with GPU-accelerated HLS transcoding under Docker Compose.
- DSA coaching platform — FastAPI and React, 150+ problems organized by concept graph rather than difficulty; 40+ students in pilot.
- Secure digital library — Node/Express API with OAuth 2.0, token rotation, and five-level RBAC; sub-200ms P95 at 800+ concurrent users.

EDUCATION

The Pennsylvania State University · M.S. Cybersecurity Operations & Technology; GPA 3.7/4.0 Aug 2024 – May 2026
Anna University, College of Engineering Guindy · Integrated M.S. Information Technology; GPA 3.3/4.0 Aug 2019 – May 2024

SKILLS

- Languages** Python, C++17, TypeScript/JavaScript, Java, Haskell, SQL, Bash, Go
- ML & Safety** PyTorch, HuggingFace Transformers, vLLM, TransformerLens, sparse autoencoders, activation patching, linear/deception probes, model evaluations, red-teaming, TensorFlow, scikit-learn
- LLM Systems** Anthropic API, MCP tool-calling, Flower AgentApp, RAG, ChromaDB, pgvector, LangChain, Ollama, llama.cpp, quantization, prompt-injection defense, grounding validation
- Security** Threat modeling, RBAC/ACL design, OAuth 2.0, identity & access management, intrusion detection, SAST/DAST, network security
- Infrastructure** AWS (Lambda, EC2, SQS, DynamoDB, CDK), Docker, Kubernetes, FastAPI, Node.js/Express, PostgreSQL, Redis, MinIO, Nginx, CUDA, Git, CI/CD, GitHub Actions
- Certifications** Cisco CCNA · CompTIA Security+ · Google AI Professional Certificate · Anthropic AI Fluency and Claude 101
- AI Safety Training** BlueDot Technical AI Safety Course · ARENA 3.0, Neel Nanda track · Agentic AI Summit 2026, Berkeley RDI
- Recognition** MATS — advanced to stage 2, empirical track · First place, 24-hour blockchain hackathon, NIT Trichy · Asia Book of Records and Limca Book of Records, five-day long-duration skating